WORKED SOLUTION

STEP I 2003, Question 1

Pure • Sums of Powers

STEP I 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q1

The key idea

We are told the shape of the answer, a polynomial in n with unknown coefficients, so all that is left is to pin down those coefficients. A polynomial identity that holds for every n certainly holds for a few convenient values of n, and each value gives one linear equation. Starting the sums at a negative index is a deliberate trick: it lets us use small values such as $n = -1$ and $n = -2$, which makes the equations pleasantly symmetric.

Solution

The question uses r both as the summation variable and as a coefficient. To avoid the clash we write the summation variable as k, keeping p, q, r, s and a, b, c, d, e for the coefficients.

Sum of squares

Let $S(n) = \sum_{k=-1}^{n} k^2$, so that

$$S(n) = pn^3 + qn^2 + rn + s.$$

The sum starting at $k = -1$ makes sense for every $n \geq -1$, and we have

$$S(-1) = 1, \qquad S(0) = 1+0 = 1, \qquad S(1) = 1+0+1 = 2, \qquad S(2) = 1+0+1+4 = 6.$$

Substituting $n = -1, 0, 1, 2$ into the given form gives the four equations

$$-p + q - r + s = 1, \tag{1}$$
$$s = 1, \tag{2}$$
$$p + q + r + s = 2, \tag{3}$$
$$8p + 4q + 2r + s = 6. \tag{4}$$

From (2), $s = 1$, so (1) and (3) become $-p + q - r = 0$ and $p + q + r = 1$. Adding these gives $2q = 1$, so $q = \frac{1}{2}$, and then $p + r = \frac{1}{2}$. Equation (4) becomes

$$8p + 2 + 2r + 1 = 6 \quad \Longrightarrow \quad 8p + 2r = 3 \quad \Longrightarrow \quad 6p + 2(p + r) = 3,$$

so $6p = 2$, giving $p = \frac{1}{3}$ and $r = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$. Hence

$$\sum_{k=-1}^{n} k^2 = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n + 1.$$

Ask yourself The question wants the sum from $k = 0$, but we have found the sum from $k = -1$. How do they differ?

They differ only by the $k = -1$ term, which is $(-1)^2 = 1$. So

$$\sum_{k=0}^{n} k^2 = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n = \tfrac{1}{6}n\left(2n^2 + 3n + 1\right) = \tfrac{1}{6}n(n+1)(2n+1),$$

as required.

Sum of cubes

Now let $T(n) = \sum_{k=-2}^{n} k^3 = an^4 + bn^3 + cn^2 + dn + e$. There are five unknowns, so we use the five values $n = -2, -1, 0, 1, 2$:

$$T(-2) = -8, \quad T(-1) = -9, \quad T(0) = -9, \quad T(1) = -8, \quad T(2) = 0.$$

This gives

$$16a - 8b + 4c - 2d + e = -8, \tag{5}$$
$$a - b + c - d + e = -9, \tag{6}$$
$$e = -9, \tag{7}$$
$$a + b + c + d + e = -8, \tag{8}$$
$$16a + 8b + 4c + 2d + e = 0. \tag{9}$$

From (7), $e = -9$. The equations come in pairs at $\pm n$, so it pays to add and subtract each pair: the even powers and the odd powers then separate.

Using $e = -9$, (6) and (8) become $a - b + c - d = 0$ and $a + b + c + d = 1$. Adding and subtracting,

$$a + c = \tfrac{1}{2}, \qquad b + d = \tfrac{1}{2}.$$

Similarly (5) and (9) become $16a - 8b + 4c - 2d = 1$ and $16a + 8b + 4c + 2d = 9$. Adding and subtracting,

$$32a + 8c = 10, \qquad 16b + 4d = 8,$$

that is, $16a + 4c = 5$ and $4b + d = 2$. Substituting $c = \frac{1}{2} - a$ gives $12a + 2 = 5$, so $a = \frac{1}{4}$ and $c = \frac{1}{4}$. Substituting $d = \frac{1}{2} - b$ gives $3b + \frac{1}{2} = 2$, so $b = \frac{1}{2}$ and $d = 0$. Hence

$$\sum_{k=-2}^{n} k^3 = \tfrac{1}{4}n^4 + \tfrac{1}{2}n^3 + \tfrac{1}{4}n^2 - 9.$$

The sum from $k=-2$ contains the extra terms $(-2)^3+(-1)^3=-9$ compared with the sum from $k=0$, so we add 9:

$$\sum_{k=0}^{n} k^3 = \tfrac{1}{4}n^4+\tfrac{1}{2}n^3+\tfrac{1}{4}n^2 = \tfrac{1}{4}n^2\left(n^2+2n+1\right) = \tfrac{1}{4}n^2(n+1)^2,$$

as required.

Checks

For $n=10$ the formulae give $\frac{1}{6}\cdot 10\cdot 11\cdot 21 = 385$ and $\frac{1}{4}\cdot 100\cdot 121 = 3025$, and adding up $0^2+1^2+\cdots+10^2$ and $0^3+1^3+\cdots+10^3$ directly gives the same numbers. Notice also that $3025 = 55^2 = (0+1+\cdots+10)^2$: the sum of the first n cubes is the square of the sum of the first n integers. We also checked by computer that both formulae hold exactly for every n from 0 to 59.

Common slips

- Forgetting the $k=-1$ (or $k=-2$ and $k=-1$) terms when passing from the sum we computed to the sum the question wants. Note that $(-2)^3+(-1)^3=-9$, not $+9$.
- Trying to use only four values of n for the cubes: five unknowns need five equations.
- Confusing the coefficient r with the summation variable r in the question; renaming one of them avoids errors.

Beyond STEP

Why may we assume the answer is a polynomial of degree one higher? Because the difference $P(n)-P(n-1)$ of a polynomial of degree $m+1$ has degree m, so one can always solve for a polynomial whose differences are n^m; this is the start of the *calculus of finite differences*, and the coefficients that appear in general are governed by the Bernoulli numbers (Faulhaber's formula). The method used here, fixing a polynomial from its values at a few points, is *polynomial interpolation*, which reappears in numerical analysis as Lagrange interpolation. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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